

Treatment-related progressive multifocal leukoencephalopathy: current understanding and future steps

Progressive multifocal leukoencephalopathy (PML) is a rare demyelinating disorder of the brain caused by a ubiquitous polyomavirus, JC virus. PML is almost always associated with some underlying immunosuppression and acquired immune deficiency syndrome has been the most common predisposing disorder. Recently, different pharmacological agents have been demonstrated to increase the risk of PML. Therapies that predispose people to PML can be classified into three categories: therapies that uniquely increase the risk for the disorder, such as the monoclonal antibodies natalizumab and efalizumab; therapies that appear to increase the risk in individuals already at risk of PML due to pre-existing conditions, such as rituximab and mycophenolate mofetil; and therapies with a mechanism of action that might suggest a potential for increased PML risk and/or with which rare cases of PML have been observed. Unlike the latter two classes, therapeutic agents uniquely increasing the risk of PML are associated with a much greater prevalence of the disorder and a latent interval from the time of drug initiation to the development of PML. PML development with pharmacological agents has provided new insight into the pathogenesis of this devastating disorder. This review focuses on the risks of PML with multiple pharmacological agents, the proposed pathogenesis with these agents, and potential risk mitigation strategies.